

Empowering Quantum Jobs: Top Five Skills For Quantum Coding That Will Be In Demand

6 min read



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Highlights

Quantum computers are getting ready to augment their classical cousins in meaningful tasks, achieving what we call 'The Quantum Advantage' While hardware evolves rapidly, there is an increasing need to develop software skills to program quantum hardware and to explore applications.

Quantum kernel developers are creating high-performance quantum circuits and have begun to develop new instructions for solving real world problems.

India has one of the largest software developer populations in the world having deep skills in cloud computing, AI and Data Science. These developers are well placed to pick up quantum computing skills.

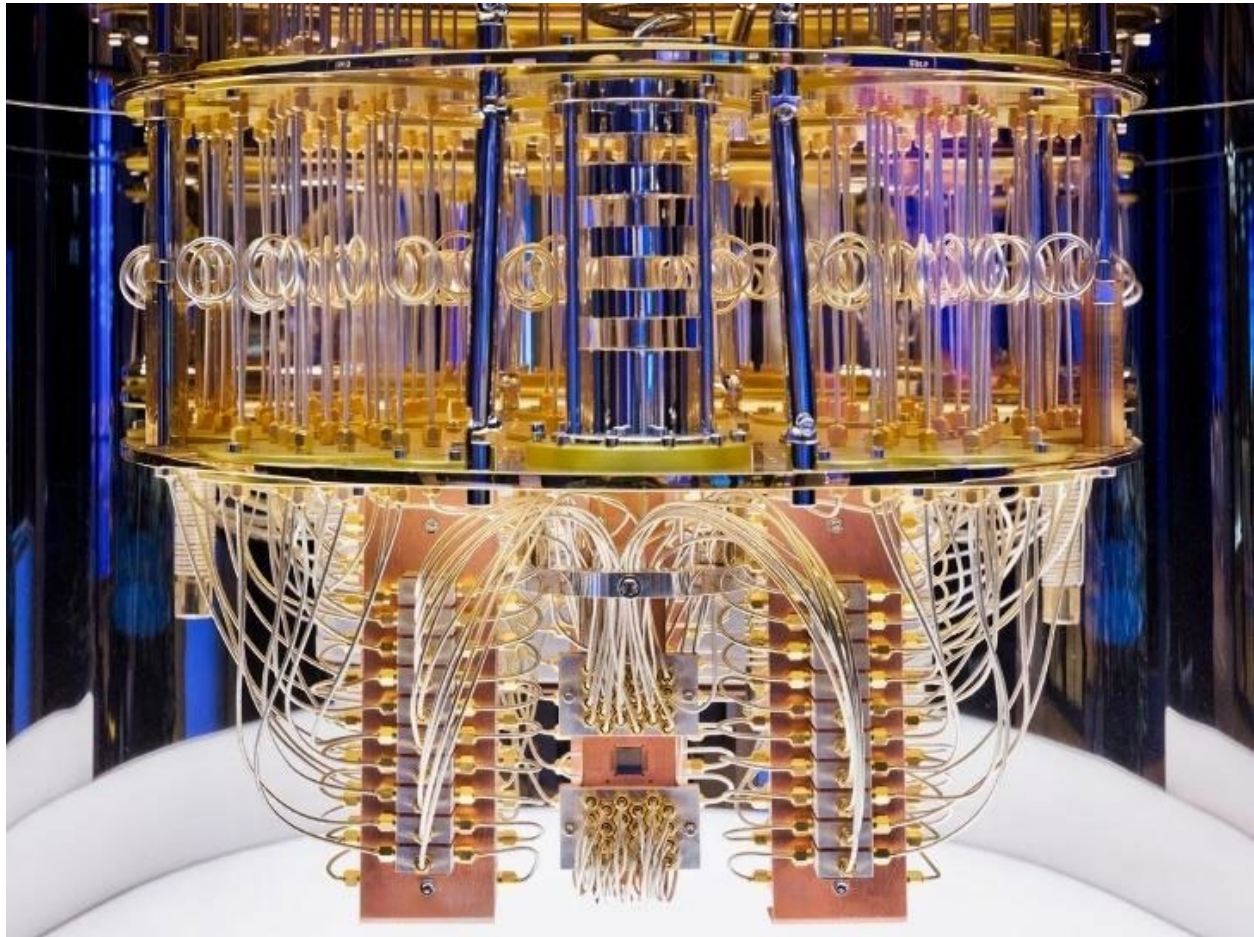
Over the past couple of years, there has been a significant interest in Quantum Computing and the areas in which this technology can be used. Quantum computing offers a great opportunity to get into an exciting new area of technology that will very likely allow humanity to harness science for overall development which will significantly impact many industries and societies.

Quantum computers are getting ready to augment their classical cousins in meaningful tasks, achieving what we call 'The Quantum Advantage'. Further, we see hardware development has been growing at exponential speed and software stack and abstractions are in place as well.

ALSO READ: [How Quantum Computing Will Help Solve Real-World Problems Much Faster](#)

While hardware evolves rapidly, there is an increasing need to develop software skills to program quantum hardware and to explore applications in areas like finance, healthcare, life sciences, natural sciences and manufacturing. Infact, a [survey from](#)

[BurningGlass Technologies](#) notes that the demand for quantum skills is expected to grow 135% in the next five years.



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The Indian government has called out quantum computing as one of the major technology disruptions that will likely change the entire paradigm of computation, communication, and encryption. It has announced a [National Mission on Quantum Technologies & Applications](#) to focus on quantum technologies. While the mission's mandate is broader, a strong component of it is to [develop skilled manpower](#) and to develop the industry ecosystem to prepare for this disruption and for India to lead it.

This is particularly pertinent for India as the [software sector](#) generates over \$175B of revenue, is nearly 8% of GDP, and employs nearly 4 million people ([investindia.gov.in](#)). Today quantum computing is at a nascent stage, and it is an opportune moment for India to take the lead and develop a strong developer workforce that is ready for the quantum era that beckons us.

Preparing for the Quantum World

The [Quantum programming](#) paradigm has gained a lot of attention in the last few years with an array of assembly formats, programming languages and software development kits. Much like in the classical world, quantum programming has now increasingly been done in high-level language with a “transpiler” that transforms and compiles to quantum hardware. An interesting part of quantum computing is that it is a technology born in the cloud and built on today’s hybrid-cloud technologies.

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This means that it is going to have to work with classical computers, use the same programming languages, but in a new design and compilation format. Quantum kernel developers are creating high-performance quantum circuits and have begun to develop new instructions for solving real world problems. The development in the Quantum field is changing rapidly; developers must be adaptable and comfortable dealing with these uncertainties.



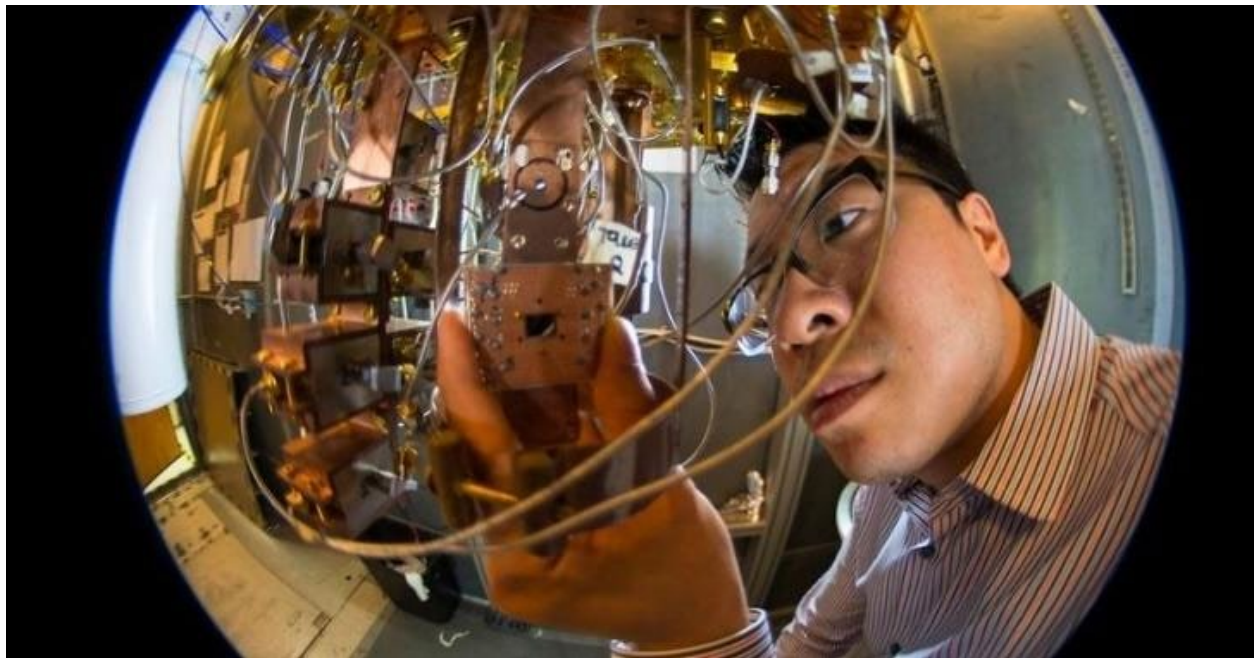
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It is important for developers to understand that quantum hardware is required only at the kernel level and the remaining software stack is largely hardware agnostic; it requires some understanding of quantum computing and deeper knowledge of algorithms and software. Developers are relying on circuits to develop groundbreaking quantum algorithms that might provide an advantage over present-day classical solutions.

Today a lot of the quantum computing software is open source, for example, Qiskit which is Python based. This means developers can easily pick up these languages and build them according to the needs of the industry domain and ecosystem of which they are a part. These Quantum models developed are used in real-world cases in automotive, finance, chemicals, medical, optimization and machine learning.

Top five skills required for Quantum coding:

1. **Programming and algorithms:** Developers should have a good grasp of languages like Python, Q#, Scaffold, among many others for quantum programming. Developers need to understand the usage of these new features in new quantum application contexts.
2. **AI/ML/DL:** Developers can potentially improve the efficiency and effectiveness of state-of-the-art quantum algorithms like quantum variational algorithms by leveraging deeper understanding of optimisers from classical AI/ML/DL contexts.
3. **Open-Source:** Quantum Software Development Kits like Qiskit are built on open-source platforms. Developers should understand how to develop, test, deploy open-source packages, design new features, or fix issues on the open-source platform.



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1. **Scientific computing:** This skill helps identify computational hard portions of the compute, which will help the developer identify the kernels of code that can potentially benefit while coding for quantum computing.

2. **Analytical reasoning and collaboration:** As Quantum is a highly interdisciplinary domain that requires working in a team with different skills, reasoning skills and collaborative spirit are sought after.

Building Quantum Skills through certification

India has one of the largest software developer populations in the world having deep skills in cloud computing, AI and Data Science. These developers are well placed to pick up quantum computing skills. As companies look to hire developers to find new quantum-powered workflows that will radically reshape their industry, quantum certifications are likely to bring about a level of standardization. Certified skills are likely to be highly valued by the industry and likely to improve career prospects significantly.

Certifications for quantum coding

One can explore the first [quantum industry developer certification](#) from IBM where they can explore several [free](#) and paid short-term online courses to advance their skills in the Quantum computing area.



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Free learning material in the form of open-source textbooks and videos are also available in [Indian languages](#).

Conclusion

Quantum computers with over 1,000 qubits are projected to be available as early as 2023 – just a couple of years away. Given the technology's disruptive potential, the industry has started mobilizing resources to grasp early learnings and start the journey to quantum.

The key to success in the journey is a strong, well-skilled, quantum workforce. For an individual developer, this is a career defining opportunity as quantum programming skills will be highly valued. For India, it is vital to grab this opportunity to secure a leadership position in Innovation that is Made in India, for India and the World.

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